

# Value-Preserving Architectures for Agentic AI Systems

Alessandro Pesare · Tommaso Dolci · Katja Hose · Emanuel Sallinger

TU WIEN · VIENNA, AUSTRIA



TECHNISCHE  
UNIVERSITÄT  
WIEN



DMKI



Knowledge  
Graph Lab



armada

BILAI



Vienna Science  
and Technology Fund

FWF

Austrian  
Science Fund

12<sup>th</sup> October 2026

# More Agents, Less Oversight

Multi-agent systems decompose goals, run in parallel and act through tools.  
Each extra agent makes behavior harder to see, attribute and control.

---

## Behavioral Opacity

Behavior cannot be understood  
from any single agent.

---

## Attributional Opacity

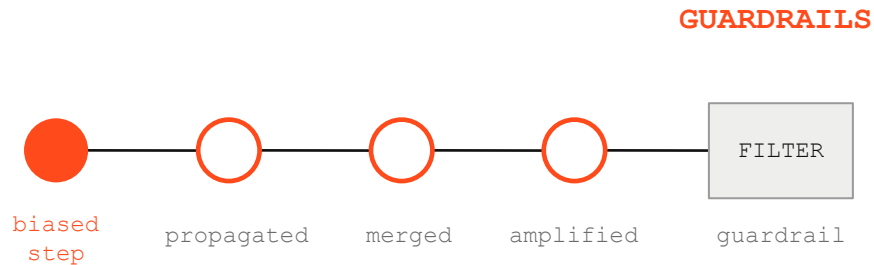
Responsibility is diluted for  
exposed records, discarded  
viewpoints, and unfairness.

---

## Control Opacity

Failure propagates silently  
within the system.

# Research Question



By the time a filter analyzes the output, a biased step or a leaked detail has already shaped the downstream steps.

## RESEARCH QUESTION

**Can we leverage architectural design to support human-centered values in the era of agentic AI?**

# Three Values, Three Architectural Patterns

VALUE

PATTERN

STRUCTURAL MECHANISM

---

**PRIVACY**

**Federated Silos Coordination**

Hub-and-spoke topology

---

**PLURALISM**

**Peer-to-Peer Deliberation**

Clique topology with no central orchestrator

---

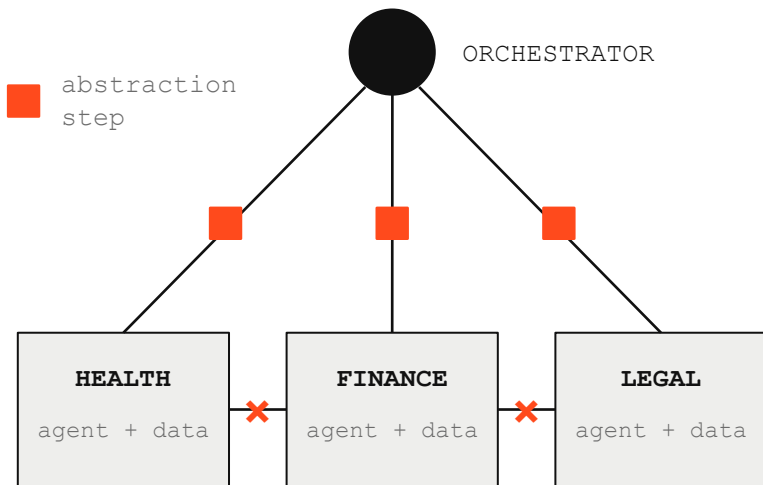
**FAIRNESS**

**Plan-Triggered Guard Agents**

Plan and review loop

---

# Federated Silos Coordination



× no inter-silo path

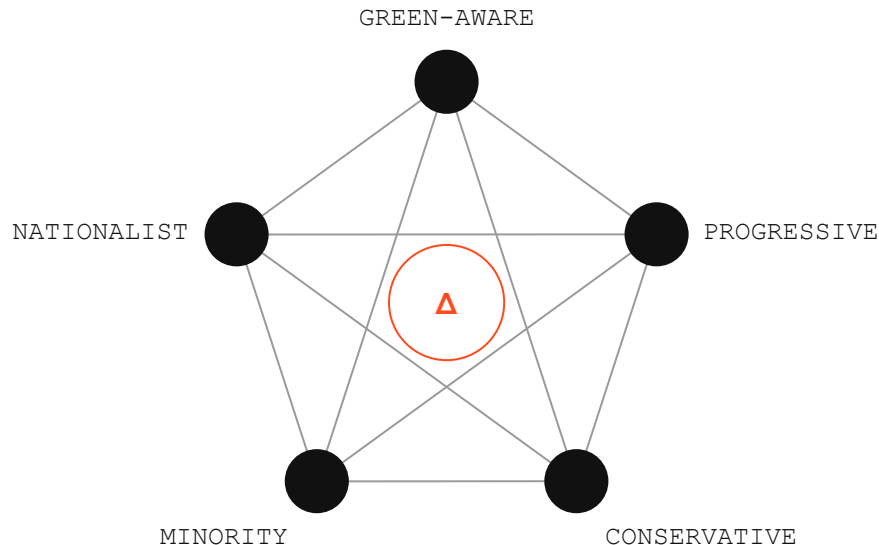
## PROBLEM

Guarantee data minimization and involve only the necessary domain agents, avoiding cross-domain communication that can expose sensitive data.

## SOLUTION

- No agent observes another **domain's records**.
- **Abstract** values before they leave the silo.
- Promote **data minimization** while retaining the information necessary to complete the task.

# Peer-to-Peer Deliberation



$\Delta$  = consensus metric - no node holds authority

## PROBLEM

Central orchestrators privilege one perspective, yet fully decentralized opinionated agents may never converge.

## SOLUTION

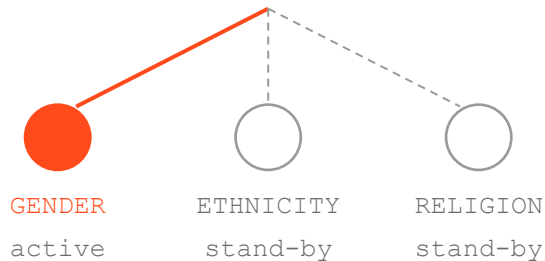
- Every agent talks **directly** to each other.
- Give each agent a distinct **deliberative position**.
- A **consensus metric** sets the stop condition and nudges agents when stalled.

# Plan-Triggered Guard Agents



no action yet

bias orchestrator



human in the loop reviewer

## PROBLEM

A single biased step can influence the downstream output, and running fairness checks at every step is not practical.

## SOLUTION

- Split the **pipeline** into plan, review, execute.
- A **bias orchestrator** inspects the plan and activates only the guard agents needed.
- **Guard agents** intervene before the plan execution.

# Use case scenarios.

## PRIVACY

### Medical treatment

Only the medical agent is activated. Debts or legal disputes never reach the orchestrator.

Treatment affordability comes from the financial agent only when asked by the user.

## PLURALISM

### News digest

A multi-agent system retrieves and deliberates over information from diverse perspectives.

Majority and minority viewpoints are equally represented.

## FAIRNESS

### Job candidate ranking

The gender guard flags a ranking that places the male candidate above a better suited female one.

The human recruiter ensures fairness in the candidate selection.

# Human-centered values are preserved from architecture.



preprint  
here!

Thank you.

alessandro.pesare@tuwien.ac.at

tommaso.dolci@tuwien.ac.at

